MIL-C-46395(MU)
5 MARCH 1964
SUPERSEDING
FA-PD-MI-2387, REV. 1
28 DECEMBER 1960

MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT, M9 ASSEMBLY

1. SCOPE

1.1 This specification covers Catapult, Aircraft Ejection Seat, M9, Assembly for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 —Ammunition and Special Weapons, General Specification for.

MIL-I-45208 —Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 —Quality Assurance Terms and Definitions.

MS-28741 —Hose Assembly—Detachable End Fitting, Medium Pressure.

DRAWINGS

MUNITIONS COMMAND

F8596866 —Catapult, Aircraft Ejection Seat, M9, Assembly.

C8826679 —Container, Ammunition, Fiber, M424 for Catapult, M9.

C8826680 —Box, Packing, Ammunition, for Catapult, M9 in Fiber Container, M424.

IEL-8596886 —Index of Inspection Equipment List for Catapult, Aircraft Ejection Seat, M9, Assembly.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Torque. Torque shall be as specified in Table I. No loosening or deformation shall occur upon application of the breakaway torque.

3.3 Residual magnetism. The assembled catapult shall not deflect the indicator of the compass more than 3 degrees in either direc-

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TABLE I.

Components	Seating or bottoming (inch-pounds)	Breakaway (inch-pounds) (minimum)
Breech, Launcher and Tube, Launcher	500+50	400
Trunnion and Tube, Launcher	400+50 (Applied between trunnion and motor tube head.)	200

TABLE II. *Booster section*

Temperature	Acceleration maximum	Rate of change of acceleration maximum (see 6.3)	Launch time maximum (see 6.4)	Acceleration-time integral minimum (see 6.5)	Catapult ignition delay maximum (see 6.2)
−65° to −70° F.	20.0g	300 g/sec	240 Millisec.	0.8 g. sec	60 Millisec.
65° to 75° F.	20.0g	300 g/sec	190 Millisec.	1.0 g. sec.	60 Millisec.
160° to 165° F.	20.0g	300 g/sec	170 Millisec.	1.2 g. sec.	60 Millisec.

TABLE III. *Rocket section*

Temperature	Pressure maximum (psi)	Pressurization rate maximum (see 6.7)	Action time maximum (see 6.8)	Pressure-time integral, minimum (see 6.9)	Rocket ignition delay maximum (see 6.6)
−65° to −70° F.	4600	100 psi/Millisec.	400 Millisec.	550 psi. sec.	50 Millisec.
65° to 75° F.	4600	100 psi/Millisec.	350 Millisec.	550 psi. sec.	12 Millisec.
160° to 165° F.	4600	100 psi/Millisec.	300 Millisec.	550 psi. sec.	12 Millisec.

tion when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 *Ballistics.* The catapult, loaded with acceptable explosive components, shall function without mechanical failure (see 6.1).

3.4.1.1 *Booster section.* The booster section shall propel a 350 pound mass and shall comply with the requirements of table II.

3.4.1.2 *Rocket section.* The rocket section shall comply with the requirements of table III.

3.5 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.5.2 *Burr.* No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies

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and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to Standard MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with Specifications MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 **Inspection provisions.**

4.2.1 *Lot.*

4.2.1.1 *Presentation of lot.* The product shall be presented in accordance with Standard MIL–STD–105.

4.2.1.2 *Size of lot.* A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of explosive components.

4.2.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in table IV. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of table IV. All nonconforming material should be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE IV.

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1— 300	100+ Examination.				
301— 800	50	0	1	1	2
801—1800	90	1	2	2	3

4.2.2.1 *Igniter, Primary Assembly, Dwg. F10524201, covering a detail of Dwg. F8596866.*

	Defect	Method of inspection
Critical:		
1.	Metal defective	Visual.
2.	Primer missing	Visual.
3.	Mixed lots of igniters	Visual.
4.	Marking missing, incorrect or illegible	Visual.
Major:		
101.	Primer pocket not sealed	Visual.
102.	Burr	Visual.
103.	Foreign matter	Visual.
Minor:	None defined.	

4.2.2.2 *Igniter, Auxiliary Assembly, Dwg. D8596874, covering a detail of Dwg. F8596866.*

	Defect	Method of inspection
Critical:		
1.	Igniter composition missing or loose	Manual/Visual.
2.	Mixed lots of auxiliary igniters	Visual.
3.	Marking missing, incorrect or illegible	Visual.
4.	Metal defective	Visual.
Major:		
101.	Burr	Visual.
102.	Foreign matter	Visual.
Minor:		
201.	Varnish missing or inadequate	Visual.

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4.2.2.3 *Holder, Strip, Loaded, Assembly, Dwg. F8596873, covering a detail of Dwg. F8596866.*

	Defect	Method of inspection
Critical:		
1.	Propellant missing or loose	Manual/Visual.
2.	Cable, ball missing or improperly assembled	Visual.
Major:		
101.	Burr	Visual.
102.	Foreign matter	Visual.
Minor:	None defined.	

4.2.2.4 *Grain, Propellant Inhibited, Dwg. F8597595, covering a detail of Dwg. F8596866.*

	Defect	Method of inspection
Critical:		
1.	End inhibitors missing or loose	Visual.
2.	Inhibited grain damaged	Visual.
3.	Wrapping loose	Visual.
4.	Voids in wrapping	Visual.
Major:		
101.	Foreign matter	Visual.
Minor:	None defined.	

4.2.2.5 *Assembly, Dwg. F8596866, less all Dwgs. except F8596873, F8596902, and Standards MS–16997–30, MS–29513–19, and MS–35337–42.*

	Defect	Method of inspection
Critical:		
1.	Improper assembly	Visual.
2.	Any part missing	Visual.
Major:		
101.	Piston slider valve not attached to holder strip on same side as propellant	Visual.
102.	Burr	Visual.
103.	Foreign matter	Visual.
Minor:		
201.	Sealing compound missing from thread	Visual.

4.2.2.6 *Assembly, Dwg. F8596866, less all Dwgs. except F8596873, D8596874, F8596893, F8596901, F8596902, C8596903, B8596908, D8596916, C8597664, A8597934, F10524201 and Standards MS–16631–1150, MS–16997–30, MS–29513–19, MS–29513–228 and MS–35337–42.*

	Defect	Method of inspection
Critical:		
1.	Improper assembly	Visual.
2.	Any part missing	Visual.
Major:		
101.	Length from face of cylinder to face of grain collar	Measure.
102.	Burr	Visual.
103.	Foreign matter	Visual.
Minor:	None defined.	

4.2.2.7 *Assembly, Dwg. F8596866, less all Dwgs. except F8596905, and Standards MS–29513–228 and MS–29513–230.*

	Defect	Method of inspection
Critical:		
1.	Improper assembly	Visual.
2.	"O" ring missing	Visual.
3.	Nozzle cups missing or loose	Manual/Visual.
Major:		
101.	Burr	Visual.
102.	Foreign matter	Visual.
Minor:	None defined.	

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4.2.2.8 *Assembly, Dwg. F8596866, less all Dwgs. except F8596873, D8596874, F10524201, F8596893, F8596901, F8596902, C8596903, F8596905, B8596908, D8596916, F8596917, D-8596919, D8597584, F8597595, C8597664, A8597934, F8597935, F8597936 and Standards MS-16631-1150, MS-16997-30, MS-29513-19, MS-29513-228, MS-29513-230, MS-35337-42 and MS-51017-35.*

	Defect	Method of inspection
Critical:		
1.	Improper assembly	Visual.
2.	Any part missing	Visual.
Major:		
101.	"O" ring damaged	Visual.
102.	Locking wire damaged, twisted or kinked	Visual.
103.	Burr	Visual.
104.	Foreign matter	Visual.
Minor:	None defined.	

4.2.2.9 *Assembly, Dwg. F8596866, less Dwgs. A8593299, D8596909, C8596910, C8596915, D8596918, F8596904. F8597936 and Standards MS-29513-12, MS-29513-18 and AN565-C8H3.*

	Defect	Method of inspection
Critical:		
1.	Cable ball not secured in holder on can	Visual.
2.	Improper assembly	Visual.
3.	Any part missing	Visual.
Major:	None defined.	
Minor:	None defined.	

4.2.2.10 *Assembly, Dwg. F8596866, less all Dwgs. except A8593299, F8596904, D8596909, C8596910, C8596915, F8597936 and Standards MS-29513-12 and MS-29513-18.*

	Defect	Method of inspection
Critical:		
1.	Improper assembly	Visual.
2.	Any part missing	Visual.
3.	Gas port obstructed	Visual.
Major:		
101.	Burr	Visual.
102.	Foreign matter	Visual.
Minor:		
201.	Shipping plug missing	Visual.

4.2.2.11 *Assembly, Dwg. F8596866.*

	Defect	Method of inspection
Critical:		
1.	Improper assembly	Visual.
2.	Labels missing, incorrect or illegible	Visual.
Major:		
101.	Centerline of lugs not aligned with mounting hole in head	Gage.
102.	Length from centerline of mounting hole to centerline of lugs	Gage.
103.	Lead seal and wire assembly missing or improperly assembled	Visual.
Minor:		
201.	Location of labels, incorrect	Visual.
202.	Varnish missing or inadequate	Visual.

4.2.2.12 *Container, Ammunition, Fiber, M424 for Catapult, M9, Dwg. C8826679.*

Critical:	None defined.	
Major:	None defined.	
Minor:		
201.	Gap between body and cover, maximum	Manual.

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	Defect	Method of inspection
202.	Contents loose	Manual.
203.	Metal ends loose or distorted	Manual.
204.	Cut or damaged through all impregnated layers	Visual.
205.	Marking missing, incorrect or illegible	Visual.
206.	Tape incomplete or badly wrinkled	Visual.
207.	Length of tear tab, minimum	Visual.
208.	Color of tape incorrect	Visual.

4.2.2.13 *Box, Packing, Ammunition, for Catapult, M9 in Fiber Container, M424, Dwg. C-8826680.*

	Defect	Method of inspection
Critical: None defined.		
Major: None defined.		
Minor:		
201.	Contents loose	Manual.
202.	Hardware missing, loose, broken or improperly engaged	Visual/Manual.
203.	Strapping missing, loose, broken or improperly assembled	Visual/Manual.
204.	Contents of box exposed	Visual.
205.	Box damaged	Visual.
206.	Marking missing, incorrect or illegible	Visual.
207.	Number of units incorrect	Visual.
208.	Car seal missing, unsealed or improperly positioned	Visual.

4.2.3 *Inspection testing.* The following tests shall be performed in accordance with 4.3, except as provided in Specification MIL–I–45208.

4.2.3.1 *Breakaway torque.* A breakaway torque shall be applied to each catapult threaded joint in 3.2. Failure of any catapult to comply with the requirement of 3.2, table I shall be cause for rejection of the catapult.

4.2.3.2 *"O" ring security.* Each subassembly consisting of valve shear pins (8596901), slider valve piston (8596902), slider valve cylinder and booster tube assembly (8596893), loaded strip holder assembly (8596873), "O" rings (MS–29513–19), cap screws (MS–16997–30) and washers (MS–35337–42) shall be subjected to the nitrogen gas test to determine that the "O" rings are properly installed and undamaged. Any subassembly showing evidence of leakage shall be cause for rejection of the subassembly.

4.2.3.3 *Residual magnetism.* Each catapult shall be tested for excessive residual magnetism. Failure of any catapult to comply with the requirements of 3.3 shall be cause for rejection of the catapult.

4.2.3.4 *Functional.* The following tests shall be performed by the Government:

4.2.3.4.1 *Ballistics.* Twenty-one units shall be tested to determine compliance with the requirements of 3.4. The units shall be temperature conditioned as provided in table V and 4.3.4.1.1. Failure of any unit to comply with 3.4 and tables II and III shall be cause for rejection of the lot.

4.2.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL–8596886, except as provided in Specification MIL–I–45208.

4.3 **Test methods and procedures.**

4.3.1 *Breakaway torque.* A torque wrench set at the torque requirements specified in 3.2, table I shall be used to determine the adequacy of the joints. The assembly shall be secured in the fixture prior to application of the torque.

4.3.2 *"O" ring security.* The subassembly shall be mounted in the device, using the fixture for properly sealing all openings. Nitrogen gas shall be injected to a pressure of 200 psi, minimum, and held for 15 seconds, minimum, to determine security of the "O" ring seals.

4.3.3 *Residual magnetism.* The catapult shall be tested for excessive residual mag-

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TABLE V.

Temperature	Sample size
—65° to —70° F.	7
65° to 75° F.	7
160° to 165° F.	7

netism in an area free of local magnetic effects by placing the catapult 6 inches from, and at the same height as, the compass in a north-south horizontal position with the compass set in an east-west heading as shown in figure 1. The catapult shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position and moved slowly back past the compass.

4.3.4 *Functional.*

4.3.4.1 *Ballistics.* The test sample shall be rigidly supported in the fixture. The sample shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this test limit is exceeded, the sample units shall be reconditioned at the original temperatures.

4.3.4.1.1 *Temperature conditioning.* A standard M9 Catapult, loaded with dummy explosive elements, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy explosive components to the required temperature.

4.3.4.2 *Method of testing.* The catapult shall be fired by an M3A1 Initiator connected to the catapult with 10 feet of aircraft hose, size 4, Standard MS–28741 to determine compliance with 3.4 and tables II and III. The head of the catapult shall be statically supported, and the weight shall be attached to the mounting lugs of the launcher and allowed to move freely. A 30 cps accelerometer shall be mounted on the mass to obtain the performance curve of the booster section. The head shall be fitted with a pressure cell to obtain the performance curve of the rocket section.

5. PREPARATION FOR DELIVERY

5.1 **Packing and marking.** (Level A) Packing and marking shall be in accordance with Drawings C8826679 and C8826680.

5.2 **Data cards.** Data card information shall be as specified in Specification MIL–A–2550.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 "Catapult ignition delay" is defined as the time that elapses between the firing of the catapult primer (t_o) and the beginning of the continuous rise (t_1) on the acceleration time record (see fig 2).

6.3 "Rate of change of acceleration" is defined as the maximum positive slope of the acceleration-time record measured over any 30 milliseconds between the continuous rise (t_1) and the maximum acceleration on the acceleration-time record (see fig 2).

6.4 "Launch time" is defined as the time between the beginning of the continuous rise (t_1) and the sudden drop (t_2) on the acceleration-time record (see fig. 2).

6.5 "Acceleration-time integral" is defined as the area under the acceleration-time record bounded by the "launch time" limits (see fig. 2).

6.6 "Rocket ignition delay" is defined as the time between the sudden drop (t_2) on the acceleration-time record of the booster section and the point at which the pressure-time record of the rocket section reaches 500 psi (t_3) (see fig. 2).

6.7 "Pressurization rate" is defined as a maximum positive slope of the pressure-time record measured over any 30 milliseconds between the point at which the pressure reaches 500 psi and the maximum pressure (see fig. 2).

6.8 "Action time" is defined as the time between the points where the pressure reaches 500 psi (t_3) at the start of burning and the

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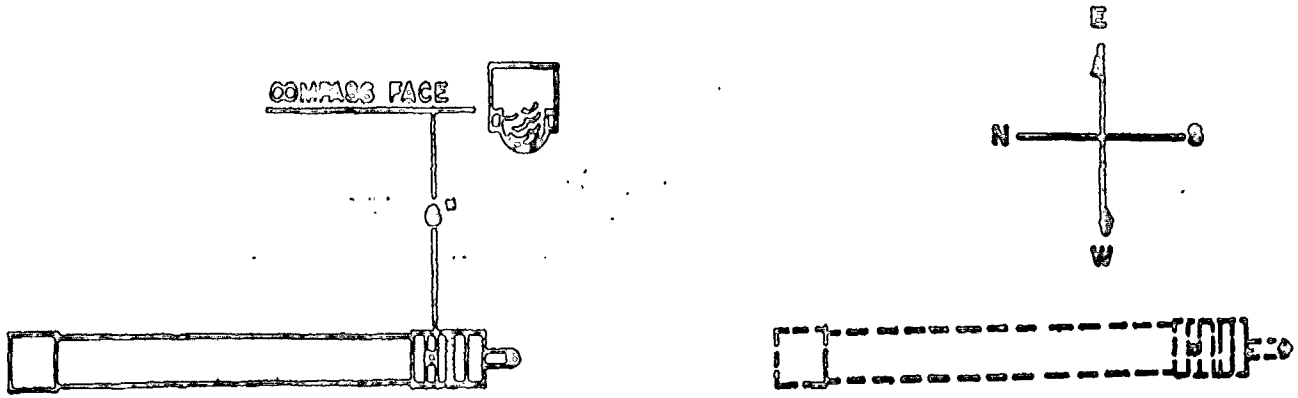
point where the pressure falls below 1000 psi (t_4) at the end of burning on the pressure-time record (see fig. 2).

6.9 "Pressure-time integral" is defined as the area under the pressure-time record bounded by the action time limits (see fig. 2).

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PLAN VIEW

NOTE:

CATAPULT AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN CATAPULT, AIRCRAFT EJECTION SEAT, M9

FIGURE 1. *Pilot's standby compass, type AN-5766.*

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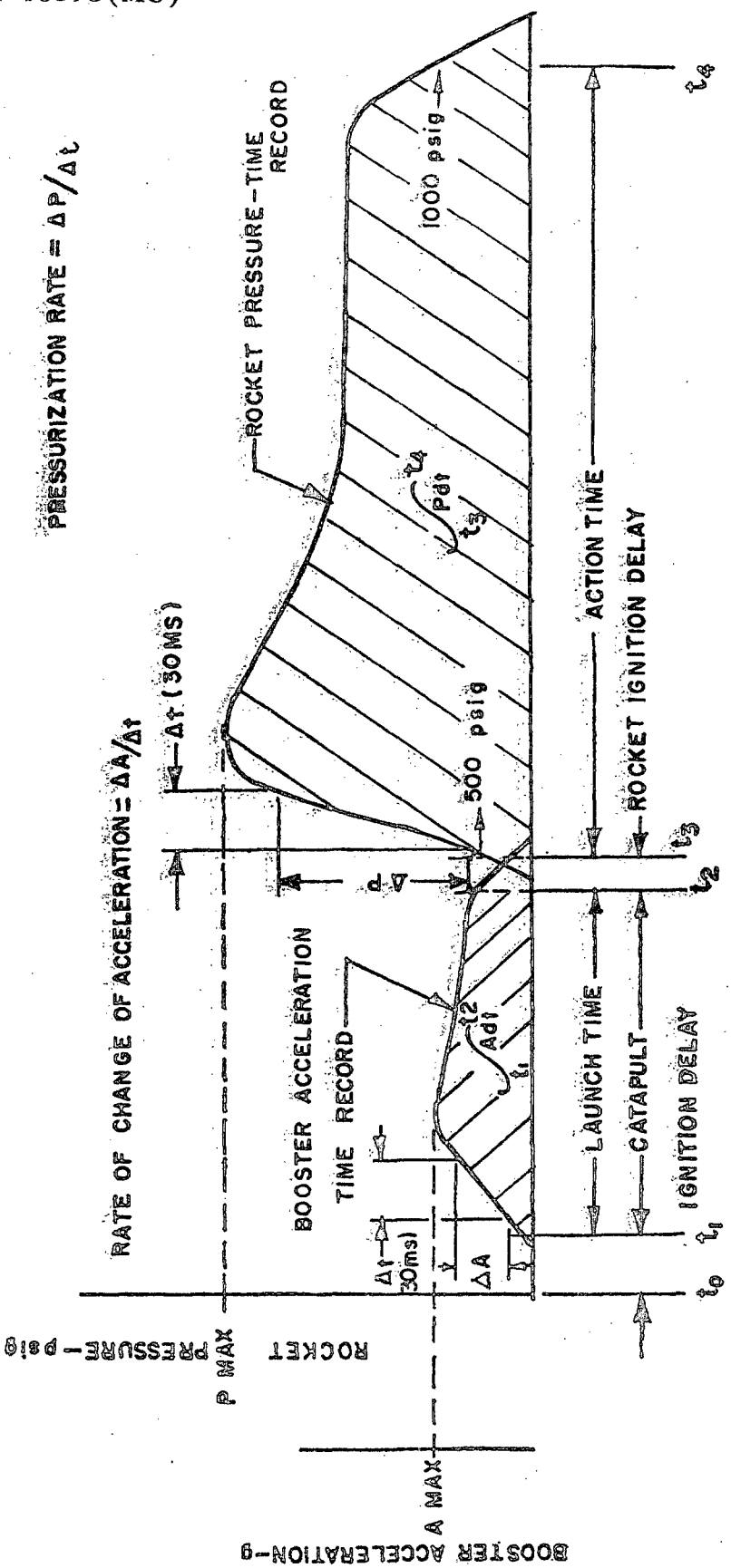


FIGURE 2. *Typical rocket catapult performance curves.*

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and mail to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY and STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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